

Structural-Thermal-Optical-Performance (STOP) analysis for the prediction of the Line of Sight stability of JANUS camera on board JUICE ESA mission

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ABSTRACT

JANUS (Jupiter Amorum ac Natorum Undique Scrutator) is a high-resolution camera to be flown on board JUICE Spacecraft, devoted to investigate the atmosphere of Jupiter and the surfaces of his icy moons (Europa, Ganymede and Callisto), in the frame of ESA “Cosmic Vision” program. The scientific objectives that JANUS will reach constrained the design of JANUS Optical Head Unit (OHU), and in particular the specific measurement of Ganymede Libration, imposes highly stringent requirement on the Line of Sight (LoS) knowledge of the instrument. The differential thermal environment conditions of the mission orbits, as well as the instrument heat dissipation timelines, induce optical elements translation and rotations that correspond to a Line of Sight variation. During the mission, the LoS can be characterized with a stellar field or single star observation but none direct measurement of its variation can be retrieved during the scientific imaging sessions. To recover the LoS knowledge, a Structural Thermal Optical and Performance Analysis (STOP) is implemented. The optics and the instrument structure are Finit Element Modeled and processed (NASTRAN) imposing the temperature distributions obtained by the OHU Thermal Model (ESATAN-TMS). The obtained thermo-elastic deformations are then considered in the OHU Optical Model (ZEMAX). The resulting LoS and the dominant OHU temperature gradients are finally entangled with a proportionality relation, as well as its direction uncertainty. The indirect estimation of the LoS variation, and its uncertainty, can be establish, thanks to STOP analysis, in real time during operations as function of installed temperature sensor measurements.

Keywords: JANUS, JUICE ESA mission, STOP analysis, Optical stability, Performances, Line of Sight, Space Instrument, High resolution camera

1. INTRODUCTION

Jupiter ICy moons Explorer (JUICE) is a mission chosen in the framework of the Cosmic Vision 2015-2025 program of the Science and Robotic Exploration (SRE) Directorate of the European Space Agency (ESA). The JUICE mission will survey the Jovian system with a special focus on the three Galilean Moons – Europa, Ganymede and Callisto – becoming the first SpaceCraft (S/C) to orbit around a giant planet Moon (Ganymede)^{1,2}. The fundamental objective of the JUICE mission is to address two key questions: *What are the conditions for planet formation and the emergence of life?* and *How does the Solar System work?* Two major science themes for JUICE have been developed: the emergence of habitable worlds around gas giants, and the Jupiter system as an archetype for gas giants. Three are the main components of the observational strategy: 1) conduct a comparative study of Ganymede, Callisto and Europa, with an emphasis on the characterization of Ganymede as a planetary object and possible habitat, 2) provide a complete spatial-temporal

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characterization of the giant, rotating magnetosphere, and of the meteorology, chemistry and structure of Jupiter's gaseous atmosphere, and 3) study coupling processes inside the Jupiter system, with a focus on the two key coupling processes within that system: the tidal effects that couple Jupiter with its satellites, and the electrodynamic interactions that couple Jupiter and its satellites with their atmospheres, subsurface oceans, magnetospheres and magnetodisc^{1,2}. The circulation, meteorology, chemistry and structure of the Jovian atmosphere will be studied from the cloud tops to the thermosphere. Within Jupiter's satellite system, JUICE will study the moons' interactions with the magnetosphere, gravitational coupling and long-term tidal evolution of the Galilean satellites^{1,2}. Remote sensing instruments include, besides the JANUS camera, the spectrometers and spectro-imagers: MAJIS, UVS, SWI.

2. JANUS

JANUS (Jovis Amorum ac Natorum Undique Scrutator) is the narrow-angle camera selected as the visible (from near UV to NIR) imager onboard JUICE^{3,4}. The JANUS instrument is composed by three units:

- Optical Head (OHU), mounted on the S/C Optical Bench
- Proximity Electronics (PEU), located close to OHU on S/C Optical Bench
- Main Electronics (MEU), located in the S/C vault

JANUS is devoted to the detailed investigation of the Galilean icy satellites, which are believed to harbor subsurface water oceans, as central objective to elucidating the conditions for habitability of icy worlds. Visible wavelength imaging is needed to determine the formation and characteristics of magmatic, tectonic, and impact features, relate them to surface forming processes, constrain global and regional surface ages, and investigate the processes of erosion and deposition⁴. JANUS will study cryovolcanism⁵ that is a plausible mechanism to explain smooth surfaces on icy bodies, including Ganymede and Europa.

In addition to the icy satellites, on which JUICE will focus during the fly-by and the final Ganymede orbit phases, the mission will allow to observe from distance (from 0.6 to few 10^6 km) all the other major target in the system, from Io and the internal satellites, to Jupiter atmosphere, rings and the other irregular satellites.

2.1 Jovian atmosphere

The main goals of JUICE with respect to the Jovian atmosphere will be related to the study of the radiative and hydrodynamic coupling. JANUS will investigate the three upper-most layers of the atmosphere via imaging observations:

- Study the troposphere weather layer, imaging the active dynamical processes, cloud systems, waves, vortices, and other instabilities, determining cloud structure in discrete features, and detecting lightning;
- Observe stratospheric variations due to vigorous water meteorology and disturbances from large vortices, such as the Great Red Spot;
- Investigate the upper atmosphere (comprising mesosphere, thermosphere and ionosphere), by imaging aurora activity and particle precipitation in the form of polar hazes.

JANUS with a resolution of 15 km/pixel observations from a 1 million km range will assure a unique view of Jupiter's ever-changing appearance of cloud structure.

In addition to cloud structure, dynamics can be directly studied by observing cloud motions over time. Wind fields are easily measurable in image pairs spanning time separations from minutes to hours.

2.2 Ganymede libration

The Galilean icy satellites are characterized by an icy crust potentially overlying a subsurface ocean with interiors displaying varying degrees of internal differentiations and processes.

Recent models suggest a variety of effects that shall be considered in order to derive information on the interior structure from bodies' libration⁶.

Librations with different periodicities but similar amplitudes are linked in different ways to the body internal structure. The capability to measure and identify these body motions by special observation plan were taken into account during the definition of the JANUS design.

3. POINTING STABILITY REQUIREMENTS

3.1 Line of Sight

The OHU Line of Sight (LoS) is defined as the angular direction coincident with the outgoing beam, with origin in the detector central pixel, nominally aligned with the optical axis. The LoS maintenance in time is called Pointing stability.

During the mission, the LoS can be characterized with respect to the S/C attitude coordinate system acquiring a stellar field or single star observation, but none direct monitoring of its stability can be retrieved during the scientific imaging sessions. Therefore is fundamental to have an opto-mechanical design devoted to minimize, during orbit, the optical elements misalignment due to thermos-elastic effects induced by operative environment. To obtain the required stability, an accurate material choice, the specific structural component geometry and a precise active thermal control strategy play a fundamental rule.

3.2 Attitude Knowledge Error

The instantaneous knowledge of the Pointing error (difference between the actual spacecraft attitude and the desired target attitude guidance reference) at any given time is called Absolute Knowledge Error of Pointing (AKE). Part of the AKE is caused by the S/C attitude control and S/C structural deformations; the remaining part is caused by the variation of the JANUS LoS.

To exploit scientific data, the LoS knowledge with associated accuracy for each image is needed whenever the instantaneous Field of View (iFoV) lacks absolute references (e.g., background stars).

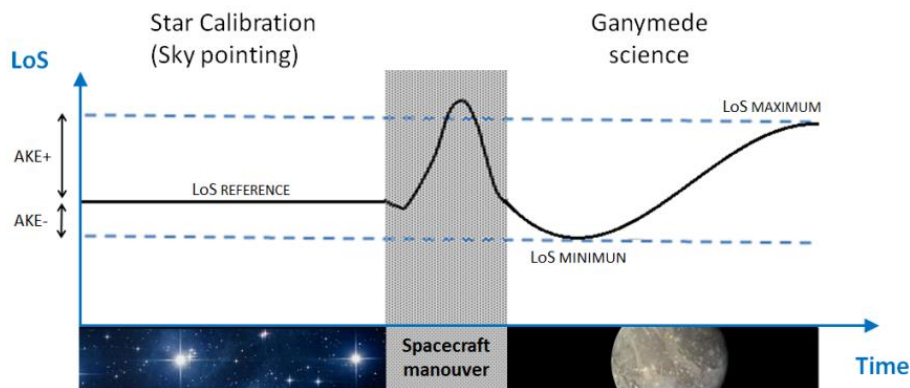


Figure 1. The LoS reference is determined during star calibration (sky pointing). Any variation from the reference LoS during the scientific acquisition around Ganymede is called AKE.

3.3 Requirements

The following operative conditions were considered by the scientific authority in order set up the pointing stability requirement driving the JANUS design:

- Nadir pointing:
 - o Ganymede orbital phases and Closest Approach (CA) during the flybys.
 - o compatibility with nominal S/C pointing.
- Out of Nadir pointing:
 - o related to stereo imaging or to specific targets (e.g., to complete target coverage, also for the rotation experiment).
 - o maximum angle of $\pm 20^\circ$ for stereo imaging during Ganymede orbit at 500 km attitude (GC0500) for Digital Terrain Model (DTM).
 - o maximum angle of $\pm 5^\circ$ for low-resolution stereo imaging and for filling gaps in global coverage during Ganymede orbit at 5000 km attitude (GC05000).

- Pointing for mosaicking:
 - o related to flyby observations (before/after the CA phase, to obtain an image mosaic of the illuminated disc of the satellites, or parts of it, during approach/departure) and Jupiter atmosphere (for global coverage, latitudinal traverses).
- Inertial pointing:
 - o related to e.g., distant objects observations and star calibration.
- Object tracking:
 - o related to special observations of Jovian cloud features, small satellites or specific event (e.g., occultation); details (slew rate and slew time) depends on the specific target.

To meet the scientific requirements the following accuracies were considered:

- a. possibility to build mosaics based on attitude information. Assuming that the majority of mosaicking shall be done for Ganymede global coverage, no binning has to be considered (images at full resolution). In this last case the needed accuracy is assumed around 1 pixels resulting in an AKE accuracy of $15\mu\text{rad}$.
- b. while observing the dynamics of Jupiter atmosphere, an AKE larger than a JANUS pixel would limit the capability to geo-locate atmospheric features during time, this results in a AKE accuracy request of $\leq 15\mu\text{rad}$.
- c. for DTM reconstruction, an AKE sub-pixel would allow obtaining the best possible absolute vertical accuracy. A higher AKE would bring to a lower absolute vertical accuracy in DTM. The result is an AKE accuracy request of $\leq 15\mu\text{rad}$.
- d. Ganymede rotation status measurements require the best possible capability for images absolute geo-location. Optimum shall be to have a sub-pixel AKE resulting in an AKE accuracy requested of $\leq 15\mu\text{rad}$.

By apportioning the global stability budget, half to S/C Optical Bench (which sustains OHU) and half to JANUS, the OHU pointing stability requirements were derived: $8\mu\text{rad}$ for GCO500 (subpixel) and $15\mu\text{rad}$ for GCO5000.

The investigation of the LoS stability, which is presented in the following chapters, only focuses on the JANUS OHU AKE contribution.

Table 1. JANUS optical requirement.

JANUS optical requirements	
Parameter	Value
iFOV	$15\mu\text{rad}/\text{pixel}$ (i.e. $7.5\text{ m}/\text{pixel}$ at GCO500)
Ground FOV	$1.72^\circ \times 1.29^\circ$
Spectral range	340-1080 nm
Optical MTF at Nyquist frequency	≥ 0.15 (at panchromatic)
Pointing stability @GCO500	$8\mu\text{rad}$
Pointing stability @GCO5000	$15\mu\text{rad}$

4. JANUS OPTICAL HEAD UNIT

To comply with the optical requirements, two main tradeoffs have determined the final configuration of JANUS OHU high-resolution camera:

- Three Mirror Anastigmatic versus Ritchey-Chrétien optical layouts – which resulted in a catadioptric telescope based on a modified Ritchey-Chrétien configuration with a dioptric corrector working in convergent beam^{7,8}.
- Invar versus Aluminum alloy and thermal control strategy of the mechanical structure – which resulted in the advantageous use of Invar having superior low thermal expansion coefficient (CTE) with respect to Aluminum, and the adoption of a single heating control loop on the structure.

Ritchey-Chrétien was the OHU optical layout chosen combined with structural elements in Invar36. OHU main architectural and functional components are:

- Filter Wheel Mechanism (FWM) – Aluminum alloy structure, black painted, with a motorization capability of switching the position of 13 broadband and narrowband Filters (designed and provided by IAA-CSIC and SENER).
- Focal Plane Module (FPM) – Titanium isostatic structure hosting the detector and a protective window made of Fused Silica (designed and provided by DLR).
- Cover Module (COM) – Mechanism protecting the OHU optical aperture from debris, molecular contaminant and possible direct Sun light exposure all along the instrument life (designed and provided by CISAS based on flight proven Rosetta mission Front Door Mechanism⁹).
- Opto-Mechanical Structure (OMS) – Invar made enclosure hosting the Telescope optics equipped with dedicated thermal control loop (to minimize in orbit development of longitudinal thermal gradients). OMS includes (designed and provided by Leonardo):
 - Telescope Module (TEM) – Invar structure with black coating to support: a Primary Mirror (M1) in Fused Silica and a Secondary mirror (M2) in Zerodur, both with hyperbolic profile and 10th order aspheric coefficient, and a triple corrector lenses in Fused Silica (L1, L2, L3) with chromatic aberration minimized.
 - Bipods – Isostatic and insulating flexible mount in Titanium.
 - Baffles – Aluminum black painted tube with vanes studied to limit first order straylight sources.
 - Connecting Wall Structure (CWS), Optical Wall (OW) and Baffle Wall (BW) – The TEM, FWM and FPM are linked to the Optical Wall (OW), while the Baffle is linked to the Baffle Wall (BW). OB and BW are stiffly linked through a Connecting Wall Structure (CWS) in a unique “box” shape.
 - Multi-Layer Insulator (MLI) – Radiative insulating blanket surrounding the whole OHU to minimize the S/C and deep space influences on the unit temperature distribution.

Table 2. JANUS geometrical and mass characteristics.

JANUS Optical Head characteristics			
Mass		Optical dimensions	
Nominal (<i>max</i>)	12.0 kg (<i>13.4 kg</i>)	Secondary Mirror to Detector	18 cm
		Primary Mirror diameter	12 cm
Envelope			
Volume	$60 \times 35 \times 25 \text{ cm}^3$		

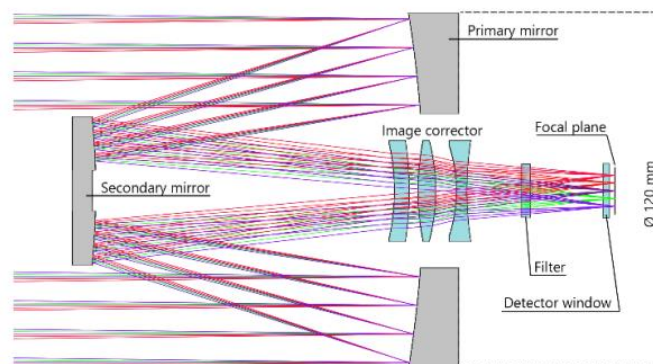


Figure 2. JANUS OHU optical layout (section along track – nadir direction).

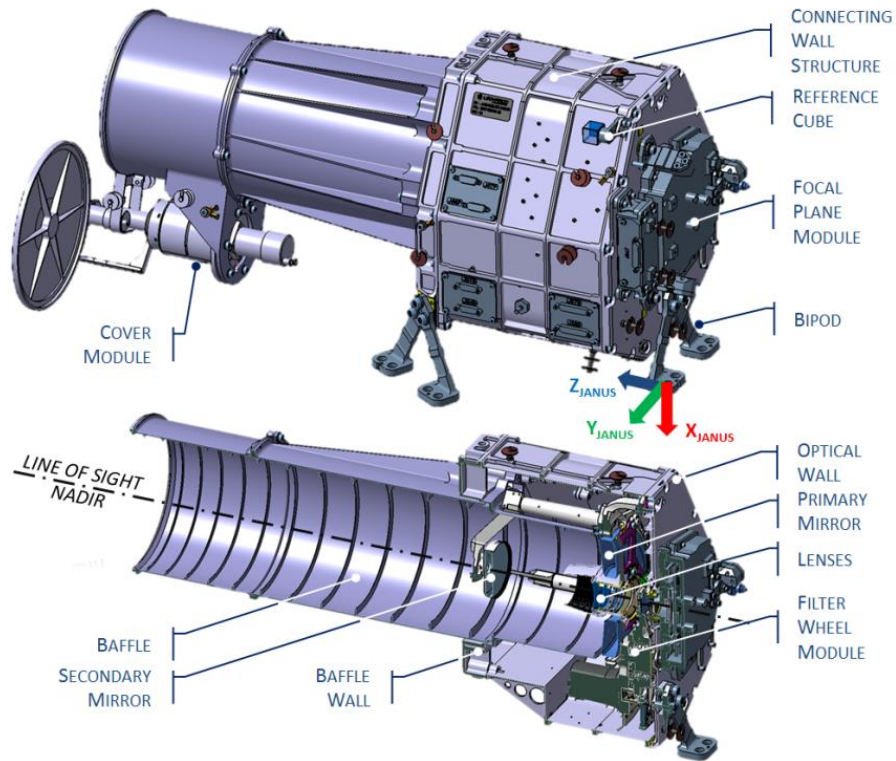


Figure 3. JANUS OHU mechanical layout (section along track – nadir direction).

The engineering actions and iterations that guided the development of the OHU opto-mechanical layout focused on the early quantification of the instrument Pointing stability and the predicted Modulation Transfer Function (MTF). The trade-offs were based on several mechanical architectures, material selections, thermo-elastic behaviors, thermal control choices and resulting optical performances.

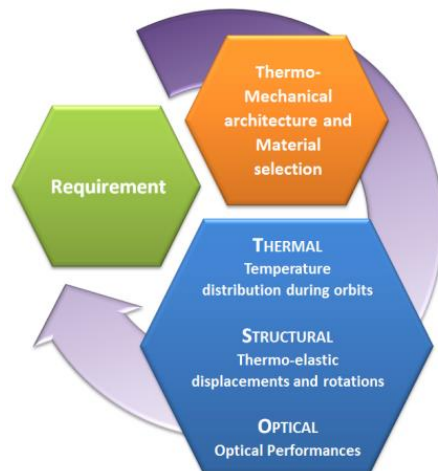


Figure 4. JANUS OHU trade-off for the selection of the thermo-mechanical architecture.

5. STOP ANALYSIS

5.1 Methodology

The orbital prediction of the AKE is computed in a sequence of multi-disciplinary analyses:

1. Thermal Analysis (ESATAN-TMS): through the OHU radiative, conductive and orbital models, the worst-case scenarios are selected in terms of temperature gradient and ambient-to-orbital temperature, and exported as temperature map distributions.
2. Thermo-Elastic Analysis (NASTRAN): the temperature map from thermal analysis is give as input to the OHU Finite Element Model, in which appropriate Coefficients of Thermal Expansion (CTE) are associated to the OHU parts. The provoked contraction/expansion generate translation and rotation of the OHU optical components (M1, M2, Lenses, Filter and Detector). The geometrical center translation and tilt of all optics and detector are exported as output.
3. Optical Analysis (ZEMAX): the translation and rotation of the optical component, obtained from thermo-elastic analysis, are imposed to the OHU Optical Model. The calculation produces the OHU LoS variation with respect the nominal assembly direction (i.e. at constant temperature in clean room condition). Degrading of the MTF is also a product of the analysis.
4. Performance Analysis: LoS and AKE uncertainty computation are extrapolated as function of the OHU temperature distribution.

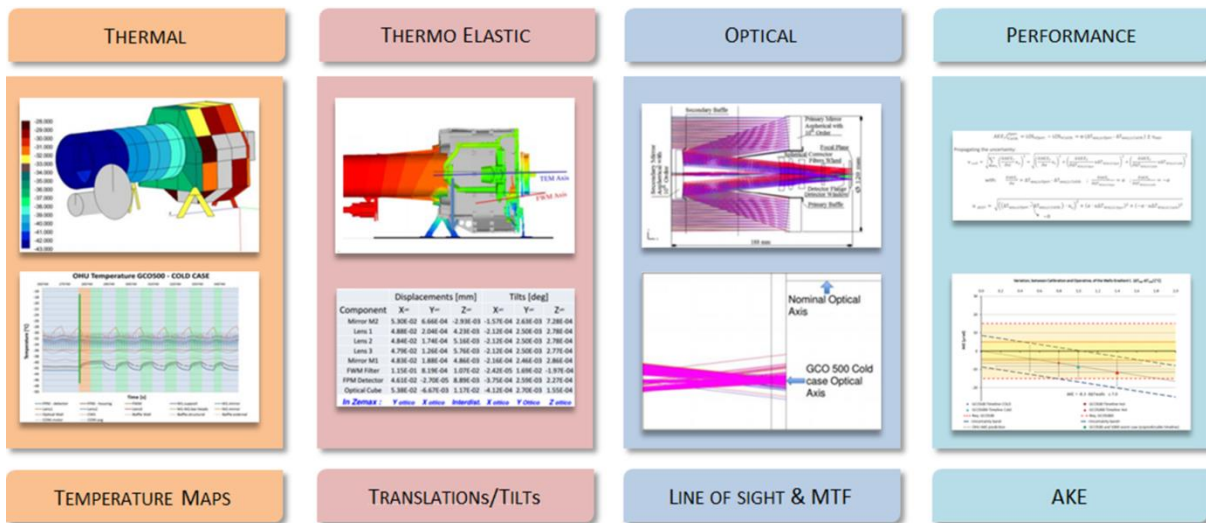


Figure 5. OHU Structural Thermal Optical Performance Analysis.

Table 3. Example of FEM derived OHU Optical Element translation and tilt in GCO500 Cold Case.

GCO500 COLD CASE OHU OPTICAL ELEMENTS DISPLACEMENT AND TILT				
Opt. Element	Displacement [μm]		Tilt [μrad]	
	$\perp$ OPT. AXIS	// OPT. AXIS	$\perp$ OPT. AXIS	// OPT. AXIS
Primary Mirror	48	5	43	5
Secondary Mirror	53	-3	46	13
Lens 1	49	4	44	5
Lens 2	48	5	44	5
Lens 3	48	6	44	5
Filter	115	11	296	-3
Detector	46	9	46	4

5.2 LoS result

Thermo-Structural-Optical analysis produces estimation of the LoS worst-case values.

Table 4. OHU LoS worst-case values.

OHU LINE OF SIGHT WORST-CASES			
GCO 500 Cold case		GCO 500 Hot case	
LoS	44 μ rad	LoS	34 μ rad
GCO 5000 Cold case		GCO 5000 Hot case	
LoS	44 μ rad	LoS	32 μ rad

5.3 LoS and AKE main contributor – Walls gradient

By comparing the results in different steps of the analysis process, the tilt value of the OMS obtained with FEM analysis (example: average value of the telescope optical elements in table 3) has a value comparable with the final Optical Model LoS (table 4), so that, with negligible difference:

$$LoS \approx TILT_{OMS} \quad (1)$$

This equivalence means that whatever is the displacement/tilt of the optical components (mirrors, lenses, and detector) inside the OMS, the major effect is caused by the global tilting of the whole OHU. In fact, when a temperature gradient (spatial) is generated between Baffle Wall and Optical Wall, the differential thermal expansion/contraction produces an OMS bending.

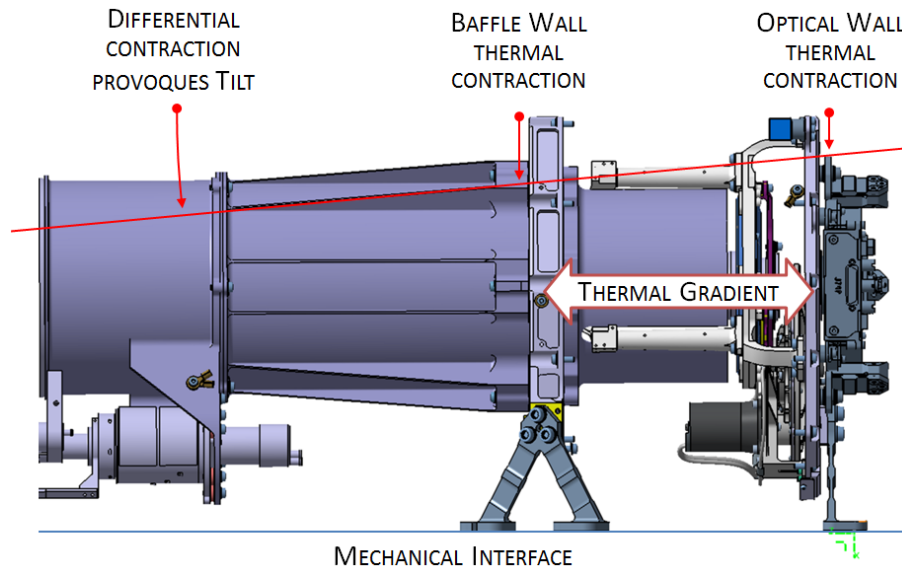


Figure 6. LoS variation due to temperature gradient (spatial) between Walls.

So, to minimize the AKE contribution virtually to zero (i.e. $LoS_{CALIBRATION} = LoS_{SCIENCE}$), any temperature gradient (spatial) between Walls shall be conserved during time (temporal derivate of the LoS equal to zero). This is called: conservation of the Walls gradient.

$$\frac{\partial \Delta T_{WALLS}}{\partial t} = 0 \text{ } ^\circ\text{C/s (i.e. conservation of the Walls gradient)} \quad (2)$$

t = Spacecraft Pointing Time-Frame Scenario (i.e. from calibration to scientific observation excluding non-operative phases).

Walls gradient influence was known and taken into consideration since from the beginning trade-off of the OHU design. To mitigate the great variation of the Walls gradient from cold to hot cases, ad-hoc location of active thermal control was studied (homogeneous distribution of heaters in the middle section of the CWS).

5.4 LoS prediction

After having deduced the physical relation between LoS and Walls gradient, the LoS can be expressed proportionally as (see also figure 7):

$$LoS \approx \alpha \cdot \Delta T_{WALLS} \pm uLoS \quad (3)$$

Note that the function intercepts the origin for null Walls gradient (once completed the optical alignment on ground, when OHU is under uniform clean room temperature condition, the LoS is nominally zero).

The whole Structural-Thermal-Optical process is affecting the LoS computation. The uncertainty sources are related to:

- OHU design geometries and material properties.
- Boundary conditions of the analysis.
- Modeling and contains assumptions.

The first two sources can be considered negligible as the OHU design is consolidated (shape, parts material characteristics, assembly components, thermo-optical properties) being the proto-flight model under manufacturing at the time of this study. Similarly the boundary conditions of the analysis (for example the conductive and radiative temperatures at S/C interfaces, the mechanical fixation location on S/C, the orbital parameters and the time lines sequence of the unit ascension) are considered invariant being requirements of the project.

So the LoS uncertainty can only be affected by Thermal-Structural-Optical modeling: But knowing that the Optical Model uncertainty is negligible with respect to Thermo-Structural uncertainty being valid equation (1) (i.e. α value is dominantly determined by structural deformation), only two sources remain:

- Thermal modeling affects the Walls Temperature gradient value.
- Structural (Finite Element) modeling affects the factor α .

The uncertainty on α and ΔT_{WALLS} are independent, in fact no parameter used in the thermal model can modify the FEM (for example the conductivity affect the temperature distribution but not the displacement, while the coefficient of thermal contraction affect the FEM displacement but not the temperature maps).

Only a material change in specific parts of the OHU might have an influence on both α and ΔT_{WALLS} , but this is already assumed to be not possible being OHU design consolidated.

So, propagating the uncertainty for independent contributors:

$$uLoS = \sqrt{\left(\frac{\partial LoS}{\partial \alpha} \cdot u\alpha\right)^2 + \left(\frac{\partial LoS}{\partial \Delta T_{WALLS}} \cdot u\Delta T_{WALLS}\right)^2} = \sqrt{(\Delta T_{WALLS} \cdot u\alpha)^2 + (\alpha \cdot u\Delta T_{WALLS})^2} \quad (4)$$

By expressing the uncertainties of the Wall gradient and of the proportional factor in percentage:

$$u\alpha = u_{\alpha\%} \cdot \alpha \quad ; \quad u\Delta T_{WALLS} = u_{\Delta T_{W\%}} \cdot \Delta T_{WALLS} \quad (5)$$

$$uLoS = \Delta T_{WALLS} \cdot \alpha \cdot \sqrt{u_{\alpha\%}^2 + u_{\Delta T_W\%}^2} \quad (6)$$

5.5 AKE prediction

The AKE is the arithmetical difference between the LoS during calibration and the LoS during science observation:

$$AKE_{Calib}^{Sci} = LoS_{Sci} - LoS_{Calib} \approx \alpha \cdot (\Delta T_{WALLS}^{Sci} - \Delta T_{WALLS}^{Calib}) \pm u_{AKE} \quad (7)$$

Where, propagating the uncertainty:

$$\begin{aligned} u_{AKE} &= \sqrt{\left(\frac{\partial AKE}{\partial \alpha} \cdot u\alpha\right)^2 + \left(\frac{\partial AKE}{\partial \Delta T_{WALLS}^{Sci}} \cdot u\Delta T_{WALLS}^{Sci}\right)^2 + \left(\frac{\partial AKE}{\partial \Delta T_{WALLS}^{Calib}} \cdot u\Delta T_{WALLS}^{Calib}\right)^2} = \\ &= \alpha \cdot \sqrt{\left((\Delta T_{WALLS}^{Sci} - \Delta T_{WALLS}^{Calib}) \cdot u_{\alpha\%}\right)^2 + (u_{\Delta T_W\%} \cdot \Delta T_{WALLS}^{Sci})^2 + (u_{\Delta T_W\%} \cdot \Delta T_{WALLS}^{Calib})^2} \end{aligned} \quad (8)$$

6. RESULTS

The proportional factor is obtained with best linear fit of the LoS STOP analysis results.

The uncertainty associated to the Walls gradient is derived following the ECSS method¹⁰ (based on sensitivities approach varying the material parameters, such as conductions, thermo-optical characteristics, heating powers, etc. within a predetermined percentage): $u_{\Delta T_W\%} = 13\%$.

The uncertainty on the proportional factor is based on engineering heritage for a FEM analysis process aiming to derive only displacements and tilts: $u_{\alpha\%} = 20\%$.

As such, the LoS and the AKE functions represented in graphical form result to be (1-sigma uncertainty):

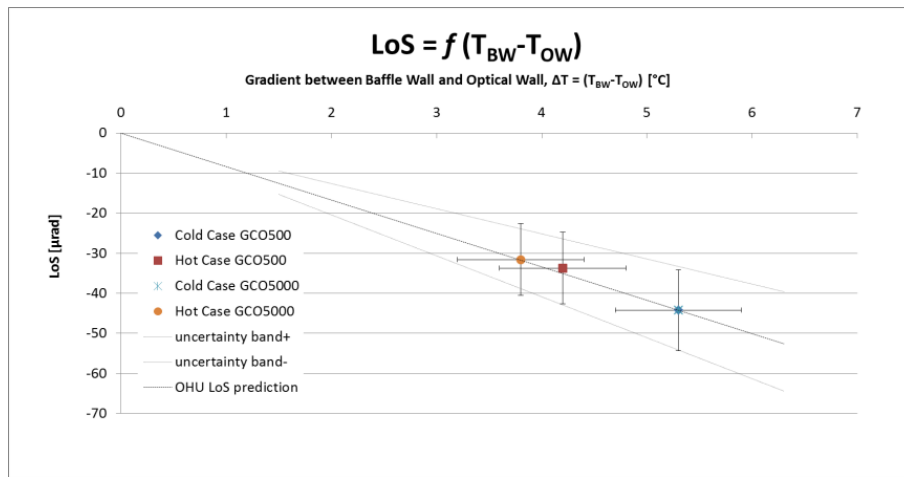


Figure 7. LoS as function of Walls gradient (1-sigma uncertainty).

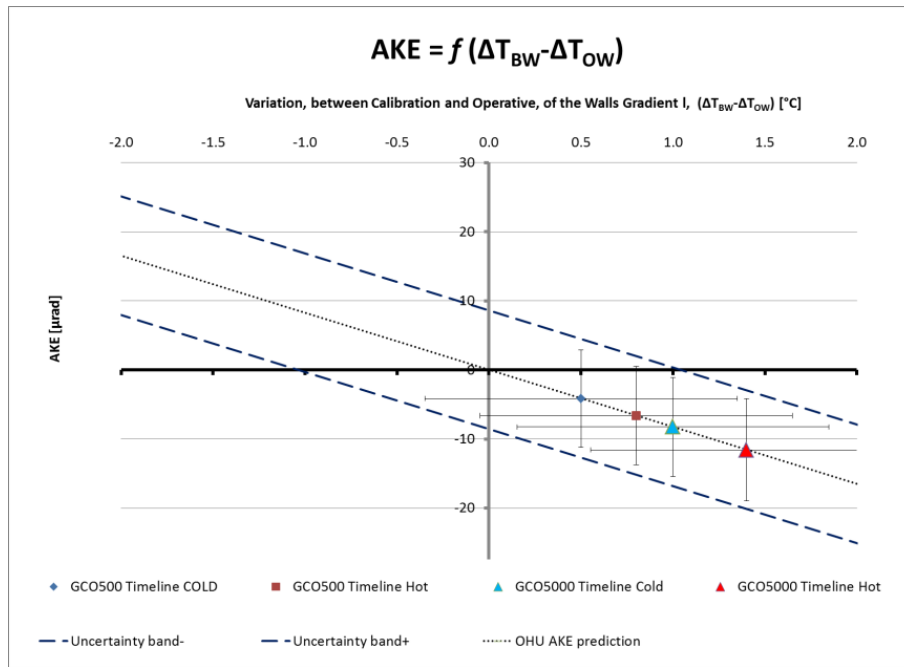


Figure 8. AKE as function of Walls gradient variation (1-sigma uncertainty).

Table 5. OHU AKE worst-case values (1-sigma).

OHU AKE WORST-CASES (1-sigma)			
GCO 500 Cold case		GCO 500 Hot case	
AKE	$-4 \pm 7 \mu\text{rad}$	AKE	$-7 \pm 7 \mu\text{rad}$
GCO 5000 Cold case		GCO 5000 Hot case	
AKE	$-8 \pm 7 \mu\text{rad}$	AKE	$-12 \pm 7 \mu\text{rad}$

7. CONCLUSIONS

JANUS Optical Head Unit, high resolution camera devoted to investigate Jupiter and its Icy Moons, is design to minimize the Line of Sight variation while orbiting (differential thermal environment conditions).

During the mission, the LoS can be characterized with a stellar field but none direct measurement of LoS variation can be retrieved during the scientific imaging sessions.

To recover the pointing error knowledge and its uncertainty, a Structural Thermal Optical and Performance Analysis (STOP) is implemented, entangling with a proportionality relation the temperature gradient of the Opto Mechanical Structure and the LoS.

Successful STOP analysis triggered the OHU hardware improvement: two dedicated temperature sensors are planned to be installed on OHU Optical Wall and Baffle Wall. Once the in orbit measurement of the Walls gradient will be transmitted with telemetry on ground, as additional information related to the JANUS image acquired, the LoS and AKE equations will allow the potential correction of the pointing error.

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